Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

ENGINEERING DESIGN AND COMPUTING

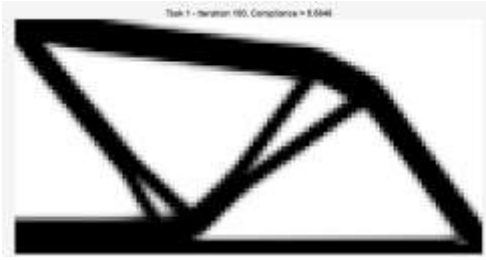
Structural Optimization 2026
Project #2 Report

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Task 1

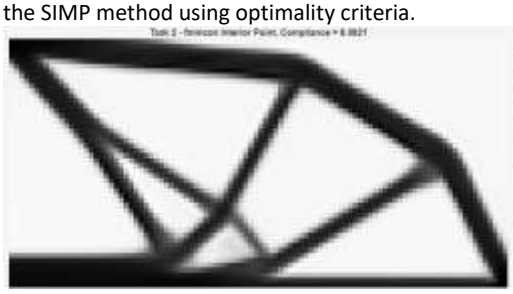
The SIMP method is implemented to minimize the compliance of the structure described in Project 2.The following parameters are used: penalty factor p = 3, move limit = 0.2, filter radius Rmin = 1.5 and max iterations = 100.The optimized structure follows the main load paths, with material concentrated along diagonal members. The best-found compliance is **5.5646 m/N** after 100 iterations.



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It.:  89 Obj.:  5.5688 Vol.:  0.349 ch.:  0.030
It.:  90 Obj.:  5.5671 Vol.:  0.350 ch.:  0.029
It.:  91 Obj.:  5.5696 Vol.:  0.351 ch.:  0.019
It.:  92 Obj.:  5.5638 Vol.:  0.350 ch.:  0.031
It.:  93 Obj.:  5.5839 Vol.:  0.350 ch.:  0.029
It.:  94 Obj.:  5.5676 Vol.:  0.349 ch.:  0.032
It.:  95 Obj.:  5.5868 Vol.:  0.350 ch.:  0.029
It.:  96 Obj.:  5.5696 Vol.:  0.351 ch.:  0.019
It.:  97 Obj.:  5.5644 Vol.:  0.350 ch.:  0.032
It.:  98 Obj.:  5.5848 Vol.:  0.350 ch.:  0.030
It.:  99 Obj.:  5.5692 Vol.:  0.351 ch.:  0.020
It.: 100 Obj.:  5.5646 Vol.:  0.350 ch.:  0.028
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Task 2

The optimization problem is solved using fmincon with the interior-point algorithm. The gradient of the objective function is computed using the adjoint method and passed to the solver. The same parameters are used as in Task 1, with a maximum of 200 function evaluations. The best-found compliance is **8.0821 m/N**.It is observed that fmincon is significantly slower than the SIMP method using optimality criteria.



```
157   189   8.101204e+00   0.000e+00   1.781e-03   3.194e-02
158   190   8.101283e+00   0.000e+00   1.776e-03   1.053e-01
159   191   8.100276e+00   0.000e+00   1.746e-03   3.019e-01
160   192   8.094842e+00   0.000e+00   1.668e-03   6.091e-01
161   193   8.085442e+00   0.000e+00   2.460e-03   7.430e-01
162   194   8.080606e+00   0.000e+00   3.245e-03   5.637e-01
163   195   8.080179e+00   0.000e+00   3.693e-03   3.253e-01
164   196   8.080152e+00   0.000e+00   3.948e-03   1.959e-01
165   197   8.080430e+00   0.000e+00   4.115e-03   1.394e-01
166   198   8.080840e+00   0.000e+00   4.239e-03   1.151e-01
167   199   8.081403e+00   0.000e+00   4.341e-03   1.063e-01
168   200   8.082128e+00   0.000e+00   4.430e-03   1.059e-01
```

Task 3

The SIMP method is used again, now considering two load cases. The objective function is defined as the sum of the compliances corresponding to each load case. The optimized structure adapts to both loading conditions and becomes more robust in regions affected by the additional load. The best-found compliance is **6.5033 m/N** after 100 iterations.



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It.:  90 Obj.:  6.5292 Vol.:  0.350 ch.:  0.027
It.:  91 Obj.:  6.5093 Vol.:  0.350 ch.:  0.016
It.:  92 Obj.:  6.5061 Vol.:  0.350 ch.:  0.013
It.:  93 Obj.:  6.5026 Vol.:  0.349 ch.:  0.022
It.:  94 Obj.:  6.5282 Vol.:  0.350 ch.:  0.027
It.:  95 Obj.:  6.5082 Vol.:  0.350 ch.:  0.017
It.:  96 Obj.:  6.5048 Vol.:  0.351 ch.:  0.013
It.:  97 Obj.:  6.5012 Vol.:  0.349 ch.:  0.021
It.:  98 Obj.:  6.5271 Vol.:  0.350 ch.:  0.027
It.:  99 Obj.:  6.5068 Vol.:  0.350 ch.:  0.017
It.: 100 Obj.:  6.5033 Vol.:  0.351 ch.:  0.012
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```

Task 4

Finally, a void region is imposed in the upper-right part of the structure, preventing material distribution in that area. The optimized structure significantly adapts to this constraint, redistributing the load paths around the excluded region. The best-found compliance is **9.3946 m/N** after 100 iterations.



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It.:  93 Obj.:  9.4149 Vol.:  0.350 ch.:  0.026
It.:  94 Obj.:  9.3945 Vol.:  0.350 ch.:  0.026
It.:  95 Obj.:  9.4150 Vol.:  0.350 ch.:  0.027
It.:  96 Obj.:  9.3946 Vol.:  0.350 ch.:  0.027
It.:  97 Obj.:  9.4151 Vol.:  0.350 ch.:  0.028
It.:  98 Obj.:  9.3947 Vol.:  0.350 ch.:  0.027
It.:  99 Obj.:  9.4151 Vol.:  0.350 ch.:  0.028
It.: 100 Obj.:  9.3946 Vol.:  0.350 ch.:  0.028
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